

PHK Early Warning Dashboard

Technical Notes

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1 Methodology for Provincial PHK Projection

1.1 Objective

The objective of the model is to project the monthly number of layoffs (PHK) at the provincial level as an early warning indicator of labour market deterioration. The model combines structural characteristics of provincial labour markets with high-frequency macroeconomic indicators to capture both long-term labour market vulnerability and short-term economic shocks.

1.2 Econometric Approach

- The dependent variable is the monthly number of reported layoffs (PHK) by province, which is a non-negative count variable with a substantial number of zero observations.
- A two-way Fixed Effects (FE) Poisson regression is employed:

$$E(PHK_{it}|X_{it}) = \exp(\beta X_{it} + \alpha_i + \gamma_t),$$

where:

- PHK_{it} denotes the number of layoffs in province i during month t ;
 - α_i represents province fixed effects;
 - γ_t represents month fixed effects;
 - X_{it} contains structural and macroeconomic variables.
- Robust standard errors are clustered at the provincial level to account for within-province serial correlation.

1.3 Rationale for Model Selection

- Provincial labour markets differ systematically in industrial composition, labour institutions, and reporting practices. These unobserved characteristics are likely correlated with macroeconomic variables, violating the assumptions required for Random Effects estimation (Hausman, 1978; Wooldridge, 2010). Fixed Effects therefore provide more consistent estimates.
- Compared with time-series approaches such as Vector Autoregression (VAR), panel Fixed Effects exploit both cross-provincial and time-series variation while accommodating relatively short monthly data and heterogeneous regional characteristics (Baltagi, 2021).
- Poisson regression is preferred because the dependent variable represents monthly counts of layoffs. Unlike OLS, Poisson guarantees non-negative predictions and naturally accommodates zero observations. Compared with Fractional Logit, which is designed for proportions bounded between zero and one, Poisson is more appropriate for count outcomes. Poisson Fixed Effects also remain consistent under relatively weak assumptions on the conditional mean even in the presence of heteroskedasticity (Hausman et al., 1984; Santos Silva & Tenreyro, 2006; Cameron & Trivedi, 2013).

1.4 Variables

Dependent Variable

- Monthly provincial PHK flow (number of reported layoffs).

Structural Variables (Annual)

- Gross Regional Domestic Product (GRDP, log)
- Manufacturing share of GRDP
- Manufacturing employment share
- Formal employment share
- Provincial minimum wage (log)

Macroeconomic Trigger Variables (Monthly)

- *Global*: Brent crude oil price.
- *National*: BI policy rate, IDR/USD exchange rate, Manufacturing PMI, Producer Price Index (IHPB).
- *Provincial*: Consumer Price Index (CPI), Non-Performing Loans (NPL), export value, and import value.

1.5 Lag Specification and Model Selection

- Macroeconomic trigger variables are evaluated using four alternative lag structures (1, 3, 6, and 12 months) to capture delayed responses of firms to economic shocks, while structural variables enter contemporaneously.
- (TBD) Candidate models are estimated using the training sample (2023–2025) and evaluated using Root Mean Squared Error (RMSE) and Mean Absolute Error (MAE) on the validation sample.
- (TBD) The specification with the lowest prediction error is selected as the preferred forecasting model and subsequently re-estimated using the full estimation sample before generating provincial and national PHK projections.

2 Results

2.1 Single-Lag Specifications

Table 1: Poisson regression estimates for provincial layoff projections

	(1) Lag 1	(2) Lag 3	(3) Lag 6	(4) Lag 12
Structure Variables				
GRDP (log)	-10.044 (8.856)	-0.034 (7.566)	-15.277 (12.476)	-19.247* (9.665)
Manufacturing Share in GRDP (%)	0.125 (0.139)	0.010 (0.144)	0.159 (0.118)	0.123 (0.104)
Manufacturing Workers (%)	-0.170 (0.137)	-0.132 (0.133)	-0.150 (0.149)	-0.194 (0.212)
Formal Workers (%)	-0.170 (0.101)	-0.158 (0.097)	-0.151 (0.108)	-0.142 (0.114)
Minimum Wage (log)	9.606 (7.183)	13.083 (8.493)	-2.816 (10.029)	13.813 (13.158)
Macroeconomic Trigger Variables				
Brent Oil Price (log)	3.003 (1.901)	0.293 (1.001)	-1.534 (1.144)	4.004*** (0.965)
BI Rate	0.138 (0.383)	0.808*** (0.219)	0.195 (0.142)	0.839* (0.370)
IDR/USD Exchange Rate (log)	-9.390 (7.045)	-0.047 (2.677)	0.209 (3.274)	-13.665* (5.590)
PMI Manufacturing	-0.114* (0.058)	-0.019 (0.069)	-0.036 (0.038)	-0.043 (0.109)
Wholesale Price Index	0.530** (0.193)	-0.071 (0.254)	0.724*** (0.173)	0.325 (0.209)
Consumer Price Index	-0.170 (0.137)	-0.074 (0.140)	-0.389* (0.175)	-0.160 (0.149)
Non-Performing Loan (log)	0.095 (0.365)	-0.248 (0.455)	-0.005 (0.631)	-0.871 (0.707)
Export (log)	-0.290 (0.235)	-0.932* (0.471)	0.065 (0.318)	0.157 (0.193)
Import (log)	-0.210* (0.083)	-0.107 (0.102)	-0.003 (0.082)	0.152* (0.075)
Observations	977	975	972	964
DV Mean	233.99	233.79	233.99	236.71
Pseudo R ²	0.582	0.581	0.584	0.591
Region FE	Province	Province	Province	Province
Time FE	Month	Month	Month	Month

Notes: The dependent variable is the monthly number of layoffs (PHK) by province. Robust standard errors clustered at the province level are reported in parentheses. Columns (1)-(4) report Poisson regression estimates using macroeconomic trigger variables lagged by 1, 3, 6, and 12 months, respectively. Annual structural variables are included contemporaneously in all specifications. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

2.2 Single-Lag Specifications - Robustness Check 1

Table 2: Poisson regression estimates for provincial layoff projections

	(1) Lag 1	(2) Lag 3	(3) Lag 6	(4) Lag 12
Structure Variables				
log_pdrb_y	-10.417 (9.177)	-0.084 (7.575)	-15.291 (12.498)	-18.707* (9.494)
manuf_pdrb_share_y	0.118 (0.138)	0.006 (0.144)	0.151 (0.125)	0.141 (0.109)
manuf_emp_share_y	-0.145 (0.161)	-0.116 (0.187)	-0.117 (0.198)	-0.335 (0.324)
formal_lab_share_y	-0.175 (0.103)	-0.160 (0.101)	-0.155 (0.108)	-0.136 (0.112)
log_wage_ump_y	9.274 (7.452)	12.916 (8.891)	-3.181 (10.648)	14.767 (13.646)
log_wage_avg_y	1.225 (2.863)	0.560 (2.816)	0.967 (2.877)	-2.822 (2.892)
Macroeconomic Trigger Variables				
L_log_price_brent	3.015 (1.885)	0.280 (1.013)	-1.542 (1.152)	4.068*** (0.944)
L_bi_rate_nat	0.131 (0.370)	0.808*** (0.220)	0.199 (0.145)	0.873* (0.363)
L_log_fx_idr_usd	-9.371 (6.993)	-0.067 (2.644)	0.133 (3.368)	-14.167** (5.397)
L_pmi_manuf_nat	-0.113* (0.057)	-0.019 (0.069)	-0.038 (0.037)	-0.041 (0.108)
L_ihpb_nat	0.529** (0.190)	-0.074 (0.246)	0.722*** (0.171)	0.327 (0.211)
L_cpi_index	-0.173 (0.137)	-0.074 (0.139)	-0.391* (0.175)	-0.167 (0.157)
L_log_npl	0.257 (0.638)	-0.187 (0.656)	0.085 (0.856)	-1.020 (0.751)
L_log_export	-0.288 (0.234)	-0.933* (0.469)	0.064 (0.316)	0.160 (0.193)
L_log_import	-0.205* (0.082)	-0.106 (0.101)	-0.005 (0.082)	0.158* (0.077)
Observations	977	975	972	964
DV Mean	233.99	233.79	233.99	236.71
Pseudo R ²	0.583	0.581	0.584	0.592
Region FE	Province	Province	Province	Province
Time FE	Month	Month	Month	Month

Notes: The dependent variable is the monthly number of layoffs (PHK) by province. Robust standard errors clustered at the province level are reported in parentheses. Columns (1)-(4) report Poisson regression estimates using macroeconomic trigger variables lagged by 1, 3, 6, and 12 months, respectively. Annual structural variables are included contemporaneously in all specifications. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

2.3 Single-Lag Specifications - Robustness Check 2

Table 3: Poisson regression estimates for provincial layoff projections

	(1) Lag 1	(2) Lag 3	(3) Lag 6	(4) Lag 12
Structure Variables				
log_pdrb_y	-10.941 (8.689)	-0.651 (6.915)	-15.639 (11.663)	-18.901* (9.322)
manuf_pdrb_share_y	0.136 (0.142)	0.025 (0.144)	0.165 (0.113)	0.147 (0.120)
manuf_emp_share_y	-0.133 (0.159)	-0.103 (0.184)	-0.107 (0.202)	-0.329 (0.320)
agri_emp_share_y	0.067 (0.120)	0.069 (0.109)	0.050 (0.117)	0.023 (0.117)
formal_lab_share_y	-0.168 (0.101)	-0.154 (0.102)	-0.151 (0.107)	-0.133 (0.108)
log_wage_ump_y	9.846 (7.800)	13.494 (9.100)	-2.868 (10.527)	14.893 (13.804)
log_wage_avg_y	1.402 (2.771)	0.789 (2.686)	1.130 (2.747)	-2.757 (2.772)
Macroeconomic Trigger Variables				
L_log_price_brent	3.028 (1.894)	0.272 (0.995)	-1.537 (1.155)	4.063*** (0.944)
L_bi_rate_nat	0.124 (0.366)	0.804*** (0.216)	0.199 (0.141)	0.870* (0.362)
L_log_fx_idr_usd	-9.274 (6.994)	-0.060 (2.646)	0.090 (3.413)	-14.123*** (5.381)
L_pmi_manuf_nat	-0.113* (0.057)	-0.018 (0.068)	-0.039 (0.038)	-0.041 (0.107)
L_ihpb_nat	0.523** (0.191)	-0.079 (0.244)	0.720*** (0.174)	0.323 (0.212)
L_cpi_index	-0.171 (0.137)	-0.073 (0.139)	-0.391* (0.173)	-0.162 (0.149)
L_log_npl	0.274 (0.642)	-0.141 (0.640)	0.137 (0.849)	-1.012 (0.740)
L_log_export	-0.309 (0.249)	-0.956* (0.485)	0.054 (0.311)	0.162 (0.195)
L_log_import	-0.204* (0.084)	-0.106 (0.101)	-0.009 (0.084)	0.157* (0.079)
Observations	977	975	972	964
DV Mean	234	234	234	237
Pseudo R ²	0.583	0.582	0.584	0.592
Region FE	Province	Province	Province	Province
Time FE	Month	Month	Month	Month

Notes: The dependent variable is the monthly number of layoffs (PHK) by province. Robust standard errors clustered at the province level are reported in parentheses. Columns (1)-(4) report Poisson regression estimates using macroeconomic trigger variables lagged by 1, 3, 6, and 12 months, respectively. Annual structural variables are included contemporaneously in all specifications. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels.

2.4 Cumulative Specifications

Table 4: Comparison of alternative lag specifications

	(1) Lag 1	(2) Lag 1+3	(3) Lag 1+3+6	(4) Lag 1+3+6+12
Structure Variables				
log_pdrb_y	-10.044	-6.462	-5.074	2.852
manuf_pdrb_share_y	0.125	0.098	0.084	-0.027
manuf_emp_share_y	-0.170	-0.218	-0.238	-0.238
formal_lab_share_y	-0.170	-0.154	-0.129	-0.152
log_wage_ump_y	9.606	4.451	-5.761	13.044
Macroeconomic Trigger Variables				
L1_log_price_brent	3.003	0.657	3.365	1.629
L1_bi_rate_nat	0.138	0.137	1.117	0.500
L1_log_fx_idr_usd	-9.390	7.774	19.332	32.150**
L1_pmi_manuf_nat	-0.114*	-0.037	-0.151	-0.332**
L1_ihpb_nat	0.530**	1.077***	-0.298	-0.802
L1_cpi_index	-0.170	-0.008	0.056	0.132
L1_log_npl	0.095	0.515	0.490	0.868
L1_log_export	-0.290	-0.079	-0.147	-0.291
L1_log_import	-0.210*	-0.243*	-0.180	-0.117
L3_log_price_brent		-0.501	4.557***	5.644**
L3_bi_rate_nat		0.968	-3.488**	-4.852**
L3_log_fx_idr_usd		0.159	14.551	7.181
L3_pmi_manuf_nat		-0.141*	-0.139	-0.196
L3_ihpb_nat		-0.894**	-0.937**	0.176
L3_cpi_index		-0.159	-0.147	-0.125
L3_log_npl		-0.757	-0.935	-1.099
L3_log_export		-0.766	-0.825	-0.942
L3_log_import		-0.089	-0.130	-0.166
L6_log_price_brent			-4.787***	2.147
L6_bi_rate_nat			2.568***	4.797*
L6_log_fx_idr_usd			22.642	2.793
L6_pmi_manuf_nat			0.063	-0.042
L6_ihpb_nat			0.826	-0.351
L6_cpi_index			-0.322	-0.411
L6_log_npl			0.207	0.730
L6_log_export			0.055	-0.067
L6_log_import			-0.001	-0.079
L12_log_price_brent				5.418*
L12_bi_rate_nat				1.021
L12_log_fx_idr_usd				-39.013**
L12_pmi_manuf_nat				-0.115
L12_ihpb_nat				0.421
L12_cpi_index				-0.293
L12_log_npl				-0.897
L12_log_export				0.054
L12_log_import				0.134
Observations	977	951	928	898
Mean monthly PHK	233.99	238.83	243.23	249.57
Pseudo R ²	0.582	0.600	0.623	0.638
Province FE	Yes	Yes	Yes	Yes
Month FE	Yes	Yes	Yes	Yes